z-lab / **Qwen3.8-27B-PARO** like 6 Follow Z Lab 1.19k

Image-Text-to-Text | Transformers | Safetensors | MLX | qwen3_5 | conversational | 4-bit precision | paroquant | arxiv:2511.10645 | License: apache-2.0

z-lab/Qwen3.8-27B-PARO

Pairwise Rotation Quantization for Efficient Reasoning LLM Inference

arXiv 2511.10645 | Blog ParoQuant | 🤗 Models

pypi v0.1.16

ParoQuant is the state-of-the-art INT4 quantization for LLMs. It closes the accuracy gap with FP16 while running at near-AWQ speed. Supports NVIDIA GPUs (vLLM, Transformers) and Apple Silicon (MLX). For more information, see https://github.com/z-lab/paroquant.

z-lab/Qwen3.8-27B-PARO is a 4-bit Qwen/Qwen3.8-27B quantized with ParoQuant. Check out other ParoQuant models from the Hugging Face collection.

Quick Start

oMLX (Apple Silicon)

Safetensors

Model size	6B params
Tensor type	I32 · F16 · I16

Chat template | Files info

MLX

Hardware compatibility Log In

4-bit MLX 18.8 GB

Inference Providers NEW

Image-Text-to-Text

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for z-lab/Qwen3.8-27B...

Base model Qwen/Qwen3.8-27B

Quantized (821) this model

On Apple Silicon, we recommend serving ParoQuant models using oMLX. Refer to the docs for more details.

Installation

```
# NVIDIA GPU (CUDA 12.9)
pip install "paroquant[vllm]"

# NVIDIA GPU (CUDA 13.0)
pip install "paroquant[vllm]" "vllm==0.19.1"
  --extra-index-url https://wheels.vllm.ai/0
  --extra-index-url https://download.pytorch

# Apple Silicon
pip install "paroquant[mlx]"
```

Interactive Chat

```
python -m paroquant.cli.chat --model z-lab/Q
```

OpenAI-Compatible API Server

For vLLM, you can directly use `vllm serve` to serve ParoQuant models:

```
vllm serve z-lab/Qwen3.8-27B-PARO --port 800
```

For other frameworks:

```
python -m paroquant.cli.serve --model z-lab/
```

For MLX, add `--vlm` if you wish to load the VLM components and use the model's multimodal features. For vLLM, VLM components are loaded by

Collection including `z-lab/Qwen3.…`

ParoQuant Collection
Pairwise … • 26 items • Upda… • 30

Paper for `z-lab/Qwen3.8-27B-PARO`

ParoQuant: Pairwise Rotation Qua…
Paper • 2511.10645 • Publi… • 14

default and can be skipped with the server argument `--language-model-only`.

> The visual components in this checkpoint is stored in original precision, and only the language components are quantized to 4 bits; as a result, the model size is larger than a fully-quantized model. Avoid loading the VLM components if you are not using the multimodal features for the best efficiency.

Docker (NVIDIA GPU)

> The following commands map the local cache directory to the container in order to persist kernel cache across runs. Remove `-v ...` to disable this behavior.

```
# Interactive chat
docker run --pull=always --rm -it --gpus all
  -v $HOME/.cache/paroquant:/root/.cache/paro
  ghcr.io/z-lab/paroquant:chat --model z-lab/

# API server (port 8000)
docker run --pull=always --rm -it --gpus all
  -v $HOME/.cache/paroquant:/root/.cache/paro
  ghcr.io/z-lab/paroquant:serve --model z-lab
```

Citation

```
@inproceedings{liang2026paroquant,
  title     = {{ParoQuant: Pairwise Rotation
  author    = {Liang, Yesheng and Chen, Haisl
  booktitle = {International Conference on Le
  year      = {2026}
}
```